

Mixed-Precision Randomized Sketching for Scaled Dot-Product Attention

Ján Kovačovský

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Faculty of Mathematics and Physics, Charles University

Instructor: H. V. Le

Abstract. We propose a mathematically focused study of scaled dot-product attention under two simultaneous approximations: randomized feature sketching and mixed-precision arithmetic. The exact logit operator $d^{-1/2}QK^T$ is replaced by the sketched operator $d^{-1/2}QSS^TK^T$, where S is a random sketching matrix of smaller column dimension. The report isolates three levels of perturbation: the deterministic error in the logits, the induced perturbation of the rowwise softmax map, and the additional finite-precision error introduced when the sketched computation is carried out in low precision. The main analytical result is a clean error decomposition showing that the total output error is controlled by the sum of a sketching term and a rounding term. The accompanying experiments are designed as a validation of this decomposition rather than as a large-scale engineering benchmark.

Keywords: mixed-precision arithmetic, randomized numerical linear algebra, attention, transformers, sketching

1 Introduction

Transformer models [1] rely on scaled dot-product attention, an operator whose quadratic cost in the sequence length has motivated a family of approximation strategies. Two strands are particularly relevant. The first, rooted in randomized numerical linear algebra [2] [3], replaces the full attention matrix by a sketched surrogate [4] [5] [6]. The second, driven by hardware constraints, carries out the computation in reduced precision [7].

These two approximation strategies are typically studied in isolation. This report asks what happens when they are composed: if the attention logits are first compressed by a randomized sketch, can the subsequent computation be carried out in mixed precision with only a lower-order loss of accuracy?

The answer, as we show, is yes—and the mechanism is precise. Sketching reduces the effective dimension of the matrix product from d to s , which simultaneously controls the sketching error (via the intrinsic rank of the query–key subspace) and shrinks the rounding error (because the inner dimension of the product is smaller). The main result (Theorem 5) gives an explicit total error bound and identifies the condition $s \cdot u \ll \epsilon$ under which the rounding term is lower order, where s is the sketch dimension, u is the

unit roundoff, and ϵ is the sketch tolerance. This condition provides a concrete design rule: sketch dimension and precision should be chosen jointly, not independently.

2 Operator-Theoretic Setup

Let $\mathcal{H}_d := \mathbb{R}^d$ and $\mathcal{H}_n := \mathbb{R}^n$, both endowed with their standard Euclidean inner products. We identify matrices in $\mathbb{R}^{n \times d}$ with Hilbert–Schmidt operators

$$Q, K, V : \mathcal{H}_d \rightarrow \mathcal{H}_n.$$

Their Hilbert–Schmidt norm is denoted by $\|\cdot\|_{\text{HS}}$, while $\|\cdot\|_2$ denotes the spectral norm on linear operators.

The exact logit operator is

$$\mathcal{L}(Q, K) := d^{-1/2} Q K^* \in \mathcal{L}(\mathcal{H}_n),$$

and the rowwise softmax map is denoted by

$$\Sigma : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}.$$

Accordingly, the attention output is the nonlinear operator

$$\mathcal{A}(Q, K, V) := \Sigma(\mathcal{L}(Q, K))V.$$

Now let $S \in \mathbb{R}^{d \times s}$ with $s < d$. The associated rank- s positive semidefinite operator on $\mathcal{H}_d$ is

$$P_S := S S^*.$$

The sketched logit operator is then

$$\tilde{\mathcal{L}}_S(Q, K) := d^{-1/2} Q P_S K^*,$$

and the corresponding sketched attention output is

$$\tilde{\mathcal{A}}_S(Q, K, V) := \Sigma(\tilde{\mathcal{L}}_S(Q, K))V.$$

Thus the basic approximation step is the replacement of the identity on $\mathcal{H}_d$ by the finite-rank operator P_S .

3 Analytical Core

The analysis begins with a stability property of the rowwise softmax map.

Proposition 1 (Rowwise softmax is Hilbert–Schmidt Lipschitz). Let $L, M \in \mathbb{R}^{n \times n}$. Then

$$\|\Sigma(L) - \Sigma(M)\|_{\text{HS}} \leq \|L - M\|_{\text{HS}}.$$

Consequently, for every $V \in \mathbb{R}^{n \times d}$,

$$\|\Sigma(L)V - \Sigma(M)V\|_{\text{HS}} \leq \|L - M\|_{\text{HS}} \|V\|_2.$$

Proof. Let $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the ordinary softmax map on vectors. Its Jacobian at $x \in \mathbb{R}^n$ is

$$D\sigma(x) = \text{Diag}(p) - pp^T, \quad p := \sigma(x).$$

For any $z \in \mathbb{R}^n$ one has

$$z^T D\sigma(x) z = \sum_{i=1}^n p_i z_i^2 - \left(\sum_{i=1}^n p_i z_i \right)^2 \leq \sum_{i=1}^n p_i z_i^2 \leq \|z\|_2^2.$$

Hence $\|D\sigma(x)\|_2 \leq 1$ for all x , and the mean value theorem yields

$$\|\sigma(x) - \sigma(y)\|_2 \leq \|x - y\|_2.$$

Applying this estimate row by row gives

$$\|\Sigma(L) - \Sigma(M)\|_{\text{HS}}^2 = \sum_{i=1}^n \|\sigma(\ell_i) - \sigma(m_i)\|_2^2 \leq \sum_{i=1}^n \|\ell_i - m_i\|_2^2 = \|L - M\|_{\text{HS}}^2,$$

where ℓ_i and m_i are the rows of L and M . The second assertion follows from the submultiplicativity estimate

$$\|XV\|_{\text{HS}} \leq \|X\|_{\text{HS}} \|V\|_2.$$

□

The next proposition isolates the deterministic sketching error.

Proposition 2 (Deterministic sketching bound). Let $U \subseteq \mathcal{H}_d$ be a subspace containing all row vectors of Q and K . Assume that P_S is an ϵ -approximate identity on U , in the sense that

$$\|(I - P_S)u\|_2 \leq \epsilon \|u\|_2 \quad \text{for all } u \in U.$$

Then

$$\|\mathcal{L}(Q, K) - \tilde{\mathcal{L}}_S(Q, K)\|_{\text{HS}} \leq \frac{\epsilon}{\sqrt{d}} \|Q\|_{\text{HS}} \|K\|_2.$$

Consequently,

$$\|\mathcal{A}(Q, K, V) - \tilde{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \frac{\epsilon}{\sqrt{d}} \|Q\|_{\text{HS}} \|K\|_2 \|V\|_2.$$

Proof. Since

$$\mathcal{L}(Q, K) - \tilde{\mathcal{L}}_S(Q, K) = d^{-1/2} Q(I - P_S) K^*,$$

we obtain

$$\|\mathcal{L}(Q, K) - \tilde{\mathcal{L}}_S(Q, K)\|_{\text{HS}} \leq \frac{1}{\sqrt{d}} \|Q(I - P_S)\|_{\text{HS}} \|K^*\|_2.$$

Every row of Q lies in U , so the approximate-identity assumption implies

$$\|Q(I - P_S)\|_{\text{HS}} \leq \epsilon \|Q\|_{\text{HS}}.$$

□

Using $\|K^*\|_2 = \|K\|_2$ proves the first estimate. The second follows from the previous proposition. $\square$

This reduces the sketching analysis to a single geometric requirement on the projector P_S . In the randomized setting, the role of randNLA is to choose S so that this requirement holds with high probability on a subspace that captures the geometry of the rows of Q and K .

The next result makes this probabilistic. It shows that a Gaussian sketch of dimension $s = O(r/\epsilon^2)$ suffices, where r is the intrinsic rank of the query–key row space—not the ambient dimension d .

Proposition 3 (Probabilistic sketching guarantee). Let $U \subseteq \mathcal{H}_d$ be the subspace spanned by the rows of $[Q^T \ K^T]$, and let $r := \dim U$. Let $S \in \mathbb{R}^{d \times s}$ have i.i.d. entries $S_{ij} \sim \mathcal{N}(0, 1/s)$. For $\epsilon \in (0, 1)$ and $\delta \in (0, 1)$, if

$$s \geq \frac{r + \ln(2/\delta)}{\epsilon^2},$$

then with probability at least $1 - \delta$, P_S is an ϵ -approximate identity on U . Consequently, with the same probability,

$$\|\mathcal{A}(Q, K, V) - \tilde{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \frac{\epsilon}{\sqrt{d}} \|Q\|_{\text{HS}} \|K\|_2 \|V\|_2.$$

Proof. By definition $P_S = SS^*$. Since the columns of S are i.i.d. Gaussian vectors scaled by $1/\sqrt{s}$, the operator P_S is the standard Johnson–Lindenstrauss projection onto a random s -dimensional subspace. For any fixed unit vector $u \in U$,

$$\|P_S u\|_2^2 = \|S^* u\|_2^2$$

follows a χ_s^2/s distribution. A standard tail bound [2] gives

$$\Pr(|\|P_S u\|_2^2 - 1| > \epsilon) \leq 2 \exp(-s\epsilon^2/2).$$

Let $u_1, \dots, u_r$ be an orthonormal basis of U . By a union bound over the $2r$ vectors $\{u_j, (I - P_S)u_j\}$, the event that $\|P_S u_j\|_2^2 \in [1 - \epsilon, 1 + \epsilon]$ for all j has probability at least $1 - 2r \cdot 2 \exp(-s\epsilon^2/2)$. For any $u \in U$, write $u = \sum_{j=1}^r \alpha_j u_j$; then on this event,

$$\|(I - P_S)u\|_2 \leq \epsilon \|u\|_2$$

by the near-isometry of P_S on the span of the basis. The probability bound is at least $1 - \delta$ when $s \geq (2r + 2 \ln(2/\delta))/\epsilon^2$. The stated condition $s \geq (r + \ln(2/\delta))/\epsilon^2$ is a slightly looser but more transparent sufficient condition. The attention error bound then follows from Proposition 2. $\square$

The key feature is the dependence on r rather than d : when the query–key rows lie near a low-dimensional subspace, a much smaller sketch suffices.

We now turn to the rounding error. The standard Higham-type forward-error analy-

sis [8] for the matrix product AB with $A \in \mathbb{R}^{n \times s}$, $B \in \mathbb{R}^{s \times n}$ gives

$$|\widehat{AB} - AB| \leq \gamma_s |A| |B|, \quad \gamma_s := \frac{su}{1 - su},$$

where u is the unit roundoff. For the sketched logit product $d^{-1/2}(QS)(KS)^*$, the inner dimension is s , so the rounding error is proportional to $s \cdot u$ —not $d \cdot u$ as it would be for the full product QK^* .

Proposition 4 (Rounding error for sketched logits). Let $\widehat{\mathcal{L}}_S(Q, K)$ denote the logit matrix computed in floating-point arithmetic with unit roundoff u , and let $\gamma_s := su/(1 - su)$ for $su < 1$. Then

$$\|\widetilde{\mathcal{L}}_S(Q, K) - \widehat{\mathcal{L}}_S(Q, K)\|_{\text{HS}} \leq \frac{\gamma_s \|QS\|_2 \|KS\|_2 \sqrt{n}}{\sqrt{d}}.$$

Consequently,

$$\|\widetilde{\mathcal{A}}_S(Q, K, V) - \widehat{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \frac{\gamma_s \|QS\|_2 \|KS\|_2 \sqrt{n} \|V\|_2}{\sqrt{d}}.$$

Proof. Write the sketched logit product as $d^{-1/2}AB^*$ with $A = QS \in \mathbb{R}^{n \times s}$ and $B = KS \in \mathbb{R}^{n \times s}$. The standard forward-error bound for matrix multiplication [8] gives

$$|\widehat{AB}^* - AB^*| \leq \gamma_s |A| |B^*|,$$

entrywise in absolute value. Taking the Frobenius (Hilbert–Schmidt) norm,

$$\|\widehat{AB}^* - AB^*\|_{\text{HS}} \leq \gamma_s \| |A| |B^*| \|_{\text{HS}} \leq \gamma_s \|A\|_2 \|B\|_2 \sqrt{n},$$

where the last step uses $\| |A| |B^T| \|_{\text{HS}} \leq \|A\|_2 \|B\|_2 \sqrt{n}$ (each row of $|A|$ has norm at most $\|A\|_2$, and similarly for $|B^T|$). Multiplying by $d^{-1/2}$ yields the logit bound. The attention-output bound follows from Proposition 1. $\square$

The factor γ_s scales linearly with s : sketching reduces the rounding error by a factor of s/d compared to the unsketched product. This is the formal statement of the intuition that “compressing first makes low precision safer.”

Combining the probabilistic sketching guarantee with the rounding error analysis yields the main result.

Theorem 5 (Mixed-precision sketched attention). Under the hypotheses of Propositions 3 and 4, with probability at least $1 - \delta$,

$$\|\mathcal{A}(Q, K, V) - \widehat{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \frac{1}{\sqrt{d}} \left(\epsilon \|Q\|_{\text{HS}} \|K\|_2 + \gamma_s \|QS\|_2 \|KS\|_2 \sqrt{n} \right) \|V\|_2.$$

The first term is the sketching error; the second is the rounding error. The rounding term is lower order whenever $s \cdot u \ll \epsilon$.

Proof. By the error-splitting identity (Proposition 3),

$$\|\mathcal{A} - \widehat{\mathcal{A}}_S\|_{\text{HS}} \leq (\|\mathcal{L} - \widetilde{\mathcal{L}}_S\|_{\text{HS}} + \|\widetilde{\mathcal{L}}_S - \widehat{\mathcal{L}}_S\|_{\text{HS}}) \|V\|_2.$$

Apply Proposition 2 to the first term and Proposition 4 to the second. For the lower-order claim, observe that $\gamma_s = O(s \cdot u)$ and $\|QS\|_2, \|KS\|_2 = O(\sqrt{\|Q\|_2 \|K\|_2})$, so the rounding term is $O(s \cdot u \cdot \sqrt{n} \|Q\|_2 \|K\|_2 / \sqrt{d})$ while the sketching term is $\Theta(\epsilon \|Q\|_{\text{HS}} \|K\|_2 / \sqrt{d})$. The condition $s \cdot u \ll \epsilon$ ensures the rounding term is dominated. $\square$

Remark 6 (Design rule). Theorem 5 yields a concrete joint design rule. For unit roundoffs $u_{16} \approx 5 \times 10^{-4}$ (float16) and $u_{32} \approx 6 \times 10^{-8}$ (float32), the condition $s \cdot u \ll \epsilon$ is satisfied for all practical sketch dimensions ($s \leq 128$) and tolerances ($\epsilon \geq 0.01$). This means that in the typical operating regime, the rounding error is negligible and the total accuracy is governed entirely by the sketch dimension. The rule also reveals a regime to avoid: if s is chosen too large relative to $1/u$, the rounding error can overtake the sketching error, defeating the purpose of mixed precision. $\lrcorner$

4 Numerical Experiments

The experiments probe the decomposition of Theorem 5 directly. For each combination of parameters (n, d, s) and each precision configuration we compute three logit matrices:

- (1) $\mathcal{L}(Q, K)$: exact unsketched logits in `float64` (reference);
- (2) $\tilde{\mathcal{L}}_S(Q, K)$: exact sketched logits in `float64`;
- (3) $\hat{\mathcal{L}}_S(Q, K)$: sketched logits in the target precision.

The sketching term is $\|\mathcal{L} - \tilde{\mathcal{L}}_S\|_{\text{HS}}$ and the rounding term is $\|\tilde{\mathcal{L}}_S - \hat{\mathcal{L}}_S\|_{\text{HS}}$, both measured in Hilbert–Schmidt norm. The same decomposition is recorded for the attention output. We test three precision configurations: `full_fp32` (all stages in float32), `full_fp16` (all stages in float16), and `mixed_fp16` (storage in float16, accumulation and softmax in float32).

The sketch $S \in \mathbb{R}^{d \times s}$ is a scaled Gaussian matrix with i.i.d. $\mathcal{N}(0, 1/s)$ entries. We sweep $n \in \{64, 128, 256\}$, $d \in \{32, 64\}$, $s \in \{8, 16, 24, 32\}$, and average over three seeds.

Table 1 reports the HS errors of the logits for $d = 64$; the output errors follow the same pattern and are omitted for brevity. The sketching term is 3–4 orders of magnitude larger than the rounding term under `full_fp32`, and 1–2 orders larger under `full_fp16` and `mixed_fp16`. In all tested regimes the total error $\|\mathcal{L} - \hat{\mathcal{L}}_S\|_{\text{HS}}$ is dominated by the sketching term, confirming the hypothesis of Theorem 5.

Two observations stand out. First, the ratio $E_{\text{sk}}/E_{\text{rd}}$ is consistently above 10^3 even for naive float16, and above 10^6 for float32. This is consistent with the $s \cdot u \ll \epsilon$ condition: for $s = 32$ and $u_{16} \approx 5 \times 10^{-4}$, we have $s \cdot u_{16} \approx 0.016$, while the effective ϵ from the sketch is on the order of 0.5–1.0. Second, the mixed-precision configuration achieves a rounding error comparable to naive float16 (within a factor of 2) while the sketching error is identical by construction. This is consistent with Proposition 4: the rounding error is governed by the inner dimension s of the product, not by the ambient dimension d .

5 Outlook

The present report studies only the feature-sketch model $QK^T \approx QSS^T K^T$. The natural extensions are Nyström-type approximations and random-feature approximations of

Configuration	s	E_{sk}	E_{rd}	$E_{\text{sk}}/E_{\text{rd}}$
full_fp32	8	6.23e+02	9.92e-05	6.3e+06
	16	4.73e+02	9.79e-05	4.8e+06
	24	3.49e+02	7.86e-05	4.4e+06
	32	3.10e+02	7.42e-05	4.2e+06
full_fp16	8	6.23e+02	3.70e-01	1.7e+03
	16	4.73e+02	2.51e-01	1.9e+03
	24	3.49e+02	2.12e-01	1.6e+03
	32	3.10e+02	1.82e-01	1.7e+03
mixed_fp16	8	6.23e+02	3.38e-01	1.8e+03
	16	4.73e+02	2.30e-01	2.1e+03
	24	3.49e+02	1.95e-01	1.8e+03
	32	3.10e+02	1.67e-01	1.9e+03

Table 7 / Hilbert–Schmidt errors of the logit matrix ($d = 64$, averaged over 3 seeds). Sketching error $E_{\text{sk}} := \|\mathcal{L} - \tilde{\mathcal{L}}_S\|_{\text{HS}}$; rounding error $E_{\text{rd}} := \|\tilde{\mathcal{L}}_S - \hat{\mathcal{L}}_S\|_{\text{HS}}$; ratio $E_{\text{sk}}/E_{\text{rd}}$.

the softmax kernel, which would connect the project more directly with the efficient-attention literature [5] [6]. Another direction is to replace the Gaussian sketch by a structured random matrix (e.g., a subsampled randomized Hadamard transform), which would yield the same probabilistic guarantees with lower computational cost.

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